

Resource bound $m = \text{ceil}(L/g_{\text{req}})$: rounding discontinuities, gap guarantee, capacity shortfall

proved symbolically (RESOURCE-CEILING-001); the shaded region marks CAPACITY_SHORTFALL

(available). Hysteresis (not shown) may hold the count within ± 1 step of the ceiling within configured margins.

Symbolically verified under stated assumptions ($L > 0$, $g_{\text{req}} > 0$)

Wolfram 15.0.0 | WorkingPrecision 50 | cross-language tol 1e-9 | $g_{\text{req}} = 2.5$ m, $m_{\text{min}} = 3$, $M = 10$

